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Combining of Stimulus Dimensions in Children's and Adults' Judgments of Area: An Information Integration Analysis

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Two experiments were conducted to study developmental changes in the integration of stimulus dimensions in an area judgment task. In Experiment 1, children 5–6 years old and adults judged 9 rectangles in a 3 (width) × 3 (height) design. In Experiment 2, 5-year-olds, 8-year-olds, 11-year-olds, and adults judged 16 rectangles in a 4 × 4 design. Following functional measurement methodology, absolute judgments on a linear graphic rating scale were obtained. Data analyses showed that subjects at all ages based their area judgments on both width and height. The algebraic rule according to which these dimensions were combined, however, changed with age. Whereas the responses of the adults followed a multiplying integration model, the 5- to 6-year-olds combined width and height additively. Between these age groups, there seems to be a gradual increase in the probability that a child will shift from an adding to a multiplying strategy. Implications of these results for possible underlying processes are discussed.

Numerous researchers of cognitive development have focused on the question of the attention a child gives to the stimulus dimensions in a given task, specifically how a child combines the information from these dimensions in the judgmental process. Perhaps the most prominent experimental paradigms relating to this issue are Piaget's conservation studies (e.g., Piaget & Inhelder, 1941). For example, in the classical test of conservation of substance, one of two identical balls of clay is transformed into a sausage shape. In order to arrive at the judgment that the amount of clay does not change with the transformation, the child must attend to both relevant dimensions (i.e., length and width) simultaneously. Moreover, a specific combination rule is necessary: Length and width have to be multiplied. According to Piaget, the logical multiplication of relations is one of the most important cognitive operations leading to a complete understanding of conservation.

Nonconservation, on the other hand, is explained by Piaget (1970) in his equilibration model as follows: The younger child considers only one dimension—length or width. If he centers on length, he will judge the sausage shape as containing more clay than the ball. If he centers on width, he will judge the quantity of clay in the sausage to have decreased under this transformation.

Whereas Piaget's interpretation of the processes underlying conservation is plausible because of the necessity of the hypothesized operations (e.g., logical multiplication), his view of nonconservation may be doubted. Recent research in other areas of cognitive development has shown that children in the typical nonconservation age do attend simultaneously in various tasks to more than one stimulus dimension (e.g., Kempler, 1971; Kosslyn, Pick, & Fariello, 1974; Odom & Corbin, 1973; Shepp & Swartz, 1976).

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Therefore, it may be speculated that the conservation failures are not primarily due to a lack of attention to the relevant stimulus dimensions but to the correct combination rule for these dimensions. It might be that children in the nonconservation age do not multiply stimulus dimensions, but that they use the somewhat simpler rule of adding. Since, in the above example, the sum of length and width of the sausage is larger than that of the ball, the use of the adding rule could easily explain the "more" responses reported by Piaget.

A general method that can be applied to test this assumption is functional measurement, a technique developed by Norman Anderson (1974a, 1974b) and his colleagues in connection with information integration theory. This research has shown that adult subjects integrate stimulus information according to simple algebraic rules, such as adding or multiplying, in a variety of judgmental tasks, ranging from psychophysics to person perception.

In contrast to the numerous studies with adults, there has been little work with children concerning the developmental aspects of information integration (Butzin & Anderson, 1973; Hendrick, Franz, & Hoving, 1975; Kun, Parsons, & Ruble, 1974; Singh, Sidana, & Saluja, 1978; Singh, Sidana, & Srivastava, 1978). In each of these experiments, relatively complex stimuli were presented. A consistent result was that children 5–6 years old did not center on one stimulus dimension but used an adding or an averaging rule to combine the dimensions. Up to now, however, no multiplicative integration has been found in this age group. Of special interest in the present context is the study of Kun et al. (1974), since they obtained a developmental trend in the integration rule. Children and adults had to predict a person's performance of a task on the basis of information about the person's ability and the effort he exerts. Whereas, in accordance with the results of the studies cited above, an adding rule characterized the responses of the youngest children (6 years), the older children and adults combined ability and effort information multiplicatively to predict performance.

The present experiments were designed to study information integration in children's and adults' judgments of the areas of rectangles. This specific task was selected because it can easily be related to the work of Piaget and his associates on quantity (Piaget & Inhelder, 1941), space (Piaget & Inhelder, 1956), and geometry (Piaget, Inhelder, & Szeminska, 1960). Further, according to the formula area = width × height, judgments of rectangle area can be considered the simplest example in which the two relevant dimensions have to be multiplied in order to arrive at a correct solution. Anderson and Weiss (1971) have shown that this model holds, in fact, for adults. Whereas it was expected in Experiment 1 that this result would be replicated for adults, it was hypothesized, on the basis of the above reasoning, that children 5–6 years old would use an adding rule in this specific task.

Experiment 1

Method

In the experimental situation, subjects judged the area of different rectangles, which were presented as "chocolate bars." Subjects were asked to imagine that the bars were broken into little pieces and that the pieces were arranged side by side to form a row. The judgment task, then, was to say how long this row would be.

Subjects. The subjects were 24 children and 10 adults. The children were recruited from two kindergartens in Frankfurt, West Germany. All of the children attending these kindergartens who were at least 5 years old were tested. Of these, 4 were rejected because they did not understand the experimental task and 1 because he did not return to the second test session. The final kindergarten sample consisted of 19 children (11 boys and 8 girls). Their ages ranged from 5.0 to 6.4 years (mean age: 5.9 years). The adult sample consisted of 1 female and 9 male subjects, all of them students or staff at the Department of Psychology of the University of Frankfurt. Their ages ranged from 23 to 38 years (mean age: 29.2 years).

Materials. The stimuli to be judged were nine rectangles that were generated from all possible combinations of three widths and three heights, of 6, 12, and 18 cm. The rectangles were cut out of brown 8-mm-thick plastic. Each of the rectangles was wrapped up in silver paper, so that it looked like a chocolate bar from the outside.

In addition to these nine experimental stimuli, two end-anchor stimuli were used, as prescribed by func-

tional measurement methodology (Anderson, 1974a, p. 232). The smaller anchor was a real chocolate bar of 4 × 4 cm, consisting of four pieces of 2 × 2 cm each. The larger anchor was a square of 20 × 20 cm. It was cut out of the same plastic as the experimental stimuli and also wrapped up in silver paper.

Judgments were made on a board 2 m long, 6 cm wide, and 2 cm thick, which lay flat at the edge of the table with the length parallel to the subject. Along the edge, opposite the subject and in view of the experimenter, was a centimeter scale.

Procedure. The children were tested as follows: The child sat, facing the experimenter, at the long side of the table on which the scale lay. At the beginning of the experiment, the experimenter gave the child the chocolate bar of 4 × 4 cm, asked him to unwrap it, break it into pieces, and form a row out of the four pieces, beginning at the left side of the scale. Having done this, the child was allowed to consume the stimulus. This introduction had a double function: to establish a good rapport and to acquaint the child with the judgment task.

The experimenter then showed the child another one of the small chocolate bars and reminded the child, without breaking the bar, that a row of its pieces would reach 8 cm from the left side of the scale. Then the child was shown the 20 × 20 cm bar and told that if this bar were broken into pieces of the same size as seen before (2 × 2 cm), the row would reach to the end of the scale (2 m).

Thereafter, the end-anchor stimuli were not referred to again. In the second phase of the experiment, the child was told that he would see other "chocolate bars" and that he should imagine that each bar was broken into pieces of the same size he had seen before and arranged side by side to form a row. The bars were laid one after another on the table in front of the child, the edges of the bars being parallel to those of the table. The child was not allowed to turn the bars.

After each presentation, the child judged the extent of an imaginary row made by laying the pieces of the chocolate bar end to end. The children indicated their judgments by pointing to a position on the board that could be read in centimeters by the experimenter.

In order to prevent elaborate arithmetical operations and the use of mediator "scales," the children were encouraged to judge as quickly as possible. After the first 3–4 stimulus presentations, in which subjects seemed to try various solution strategies, judgments generally were made spontaneously in about 4 sec on an average.

The rectangles were presented four times in a random permutation order, with the only restriction that never more than two rectangles of the same area or shape appeared in succession. The 9 × 4 = 36 stimulus presentations were divided into two sessions of 18 judgments each, separated by a 3-day interval.

The experimental procedure for the adults was in principle the same as for the children. The main difference was that the chocolate was not broken into pieces at the beginning of the session; otherwise, the presentation of the end anchors paralleled that used with the children. Furthermore, in order to clarify the childlike structure of the experiment, the adults were told that their judgments were necessary as standards against which the data collected with children could be compared. Finally, all 36 judgments were obtained in one session.

Results

Graphical analysis. The mean judgments are shown in Figure 1 for children and adults separately. The data are plotted as a function of rectangle width, with one curve for each value of height.

According to the multiplying model (Area = Width × Height), the set of curves should form a diverging fan of straight lines, with the slope of each line being proportional to the value of the rectangle height (Anderson, 1974a, p. 260). As can be seen from the right panel of Figure 1, the data of the adults follow this expected form of the curves very closely. Moreover, their mean judgments coincide with the "actual" lengths of the chocolate rows, which would be 18, 36, 54; 36, 72, 108; 54, 108, and 162 cm, respectively.

Contrary to this multiplicative pattern, the lines in the graph for the children appear largely parallel. Overall parallelism, which would correspond to an additive model, seems to be violated only by the mean judgment for the largest rectangle. This 1-point discrepancy can be explained as follows: This rectangle (18 × 18 cm) was the only one that was very similar to the one used as the upper end-anchor stimulus at the beginning of the experiment (20 × 20 cm). Most of the children thought it was the same and because they remembered that the large bar reached to the end of the scale, they judged this rectangle as being "2 m long." It should be noted that even three adults showed the same effect, which may explain the slight upward curvature of the top curve in the right panel of Figure 1.

Statistical analysis. Analysis of variance can be used for a statistical test of an adding or a multiplying model. For the multiplying model, the Width × Height interaction should, of course, be significant. Moreover, this interaction should be concentrated in its bilinear component, and the

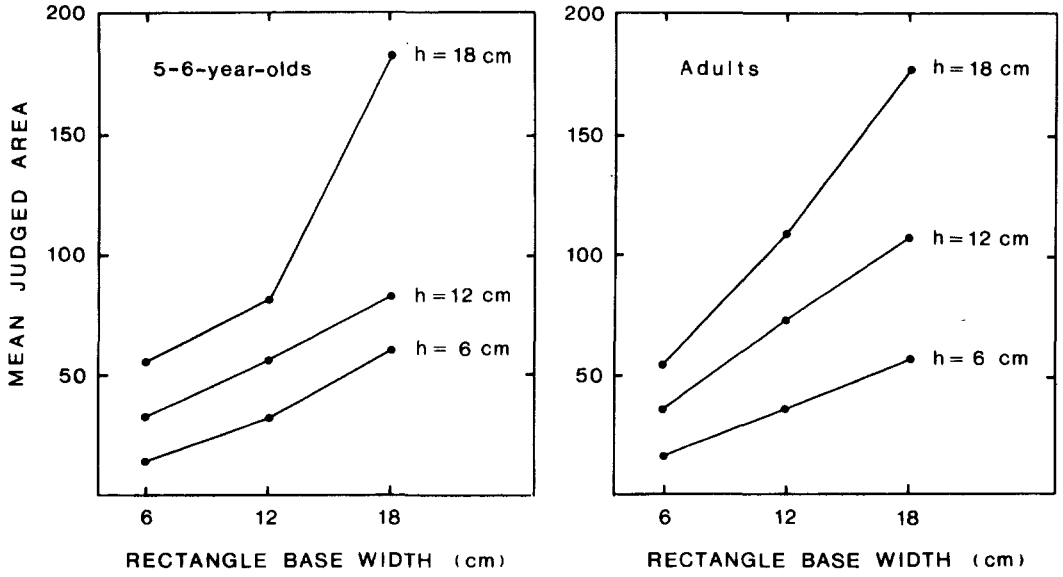

Figure 1. Mean graphic rating of area of rectangles for the children and adults in Experiment 1. Rectangle width is shown on the horizontal axis, with one curve for each value of the height (h) factor.

residual interaction should be nonsignificant (Anderson, 1974a, p. 262). In the present experiment, the analysis of variance for the adults' judgments yielded a highly significant bilinear component of the interaction, $F(1, 9) = 387.26, p < .01$. Although the total interaction variance was concentrated nearly exclusively (97%) in the bilinear part, the residual component reached significance at the .05 level, $F(3, 27) = 3.56$. An inspection of Figure 1 shows that this deviation from bilinearity is apparently caused by the slight upward curvature of the top curve noted above, that is, by the three adults who did not discriminate the largest rectangle from the end anchor. For the data of the children, on the other hand, the bilinear and the residual part of the interaction were highly significant, $F(1, 18) = 138.04$ and $F(3, 54) = 6.45$, $p < .01$, respectively (for details on computational procedure, see Graesser & Anderson, 1974).

This interaction pattern rules out a multiplying model for the children's judgments. As noted above, in this age group deviation from parallelism (i.e., from an additive pattern) seems to be caused only by the mean judgment for the largest rectangle, in which extraneous effects may be involved. Ignoring this point and calculating two analyses of variance for the two smaller values of width or height, no significant interaction was found, $F(2, 36) = .93$ and 2.33 for the 3×2 and 2×3 design, respectively. As a control, the corresponding analyses were computed for the adults. A significant interaction was still obtained for these subsets of stimuli, $F(2, 18) = 29.96$ and 38.09, $p < .01$, respectively.

In the analyses of variance for the children's judgments, main effects were always significant at the .01 level, $F(2, 36) = 79.27$ and $F(1, 18) = 60.55$ for width, and $F(1, 18) = 58.04$ and $F(2, 36) = 83.51$ for height, respectively. This result, together with the absence of interaction when the large rectangle is deleted from the analysis, suggests that the judgments of the children are best described by an additive model. Children do attend to both width and height, but, contrary to the adults, they combine the subjective values of these dimensions additively.

An additional result may serve to illustrate this finding. According to the adding model, rectangles with the same sum of edge lengths are judged as having the same area. In the present experiment, there were three rectangles with the same sum of edge lengths (6×18, 12×12, and 18×6 cm), the

square, however, being a third larger in area than the two rectangles (144 vs. 108 cm^2). As can be seen from Figure 1, children in fact did not judge these three rectangles as being different in area, $F(2, 36) = 3.03$, $p > .05$, whereas adults clearly did, $F(2, 18) = 40.43$, $p < .01$.

Experiment 2

The main purpose of this experiment was to investigate in a more precise manner the developmental shift from the adding strategy used by 5- to 6-year-olds to the multiplying strategy used by adults. To this end, two intermediate age groups were included. Furthermore, the stimulus sampling was extended, and an attempt was made to bypass the undesirable end-anchor effect that occurred in the former experiment.

Method

Subjects. The following four age groups were tested: 5-year-olds (*M* age: 5.5), 8-year-olds (*M* age: 8.5), 11-year-olds (*M* age: 11.6), and adults (*M* age: 26.4). Each age group consisted of 10 subjects, 5 male and 5 female. The 5-year-olds were recruited from a kindergarten in Frankfurt, West Germany. The 8- and 11-year-olds were attending a primary and a secondary school, respectively, in the Frankfurt area. The adults were students at the University of Frankfurt.

Materials. The materials were the same as in Experiment 1, except for the following modifications: (a) Judgments were based on a 4 × 4 design with rectangle widths and heights of 4, 8, 12, and 16 cm on each dimension, (b) no upper end anchor was used, (c) the scale was shortened to 150 cm.

Procedure. The experimental procedure paralleled that of Experiment 1 except that (a) no upper end anchor was presented; (b) in order to make the reference to the scale more difficult, the rectangles to be judged were shown vertically, held at the subject's eye level; (c) four practice trials were included using the rectangles with widths and heights of 8 × 8, 12 × 8, 4 × 8, and 16 × 12 cm, in that order; (d) following this preliminary practice, each subject judged the 16 rectangles in only two stimulus replications, which, for the 5-year-olds, were separated by a 2-day interval. For the 8- and 11-year-olds as well as for the adults, the two replications appeared in succession in one single session.

Results

Graphical analysis. Figure 2 shows the mean judgments for the four different age groups separately. The overall pattern for the 5-year-olds and the adults resembles those of the corresponding age groups in Experiment 1 very closely, indicating an adding and a multiplying model, respectively. For the 5-year-olds, the parallel (i.e., additive) pattern is even clearer than in Experiment 1, a consequence of the disappearance of the 1-point discrepancy, which was explained as an end-anchor effect above. For the adults, the omission of the end anchor brought about utilization of the entire scale, resulting in a slight overestimation of most rectangles. This overestimation did not, however, affect the clear multiplicative pattern.

The interpretation of the graphs for the two intermediate age groups is not so straightforward, for their judgment patterns cannot as unequivocally be associated

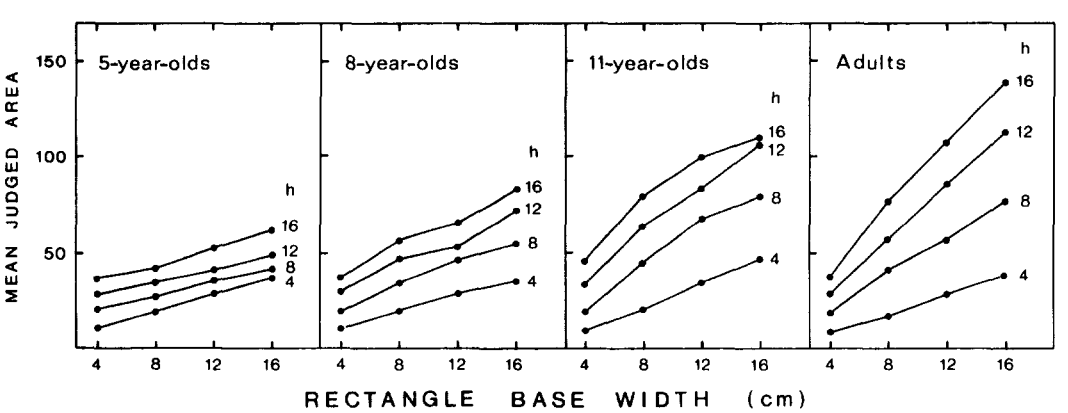

Figure 2. Mean graphic rating of area of rectangles for the four different age groups in Experiment 2. Rectangle width is shown on the horizontal axis, with one curve for each value of the height (h) factor.

Table 1
Number of Subjects Using an Adding or Multiplying Integration Rule as a Function of Age

Age group	n	Integration model	
		Adding	Multiplying
5-year-olds	10 (1)	8	1
8-year-olds	10 (2)	5	3
11-year-olds	10 (2)	3	5
Adults	10 (0)	0	10

Note. Model classifications are based on an inspection of individual graphs. Numbers in parentheses indicate number of subjects with equivocal judgment patterns.

with one of the typical integration models. On the whole, it appears as if in the course of development the adding model gradually "unfolds" into a multiplicative one.

In order to understand this unfolding in more detail, a single-subject analysis seems to be useful. An inspection of the individual graphs revealed that most of them could relatively easily be classified as either an additive or a multiplicative pattern. The results of this graphical analysis for single subjects appear in Table 1.

It can readily be seen that the number of subjects using an adding rule decreases with age, whereas the number of subjects using a multiplying rule increases. Further, the data in Table 1 show that for the 5-year-olds and the adults, the group graphs of Figure 2 reflect the individual data quite well. For the 8- and 11-year-olds, on the other hand, the group data seem to obscure different individual strategies.

Statistical analysis of integration rule. A two-way factorial analysis of variance was computed for each age group separately: For the 5-year-olds, main effects for width and height were highly significant, $F(3, 27) = 37.88$ and 40.92, $p < .01$, respectively. The interaction between the two dimensions was not significant, $F(9, 81) = .75$. Thus, the data replicate the adding integration rule obtained for the 5- to 6-year-olds in Experiment 1.

For the 8-year-olds, main effects were also significant, $F(3, 27) = 62.37$ and 63.13, $p < .01$, for width and height, respectively. The interaction $F(9, 81)$ increased to 2.35, reaching significance at the .05 level. Decomposition of this interaction into its bilinear and residual components shows that 78% of the interaction variance is accounted for by the bilinear component, $F(1, 9) = 16.51$, $p < .01$, whereas the residual is not significant, $F(8, 72) = .58$. Thus, a multiplying model may be accepted as best fit to the group data of the 8-year-olds. It should be noted, however, that the significant bilinear (i.e., multiplicative) component is caused only by the judgment patterns of 3 out of 10 children. The conclusion is that the decision on the group of the 8-year-olds as a whole is somewhat unclear, which is easily understood by considering the individual data of Table 1.

For the 11-year-olds, main effects for width and height, $F(3, 27) = 94.20$ and 76.56, respectively, as well as the overall interaction, $F(9, 81) = 4.74$, were significant at the .01 level. Although the bilinear part accounted for 59% of the total interaction variance, $F(1, 9) = 25.30$, $p < .01$, the residual part was also significant, $F(8, 72) = 2.98$, $p < .01$. The significance of the residual, as well as the relatively low bilinear component, is apparently caused mainly by one very strange data point of one of the two children with unclassifiable judgment patterns (cf. Table 1). This interaction pattern implies that the group data of the 11-year-olds, on the whole, follow neither an adding nor a multiplying model.

For the adults, the analysis of variance yielded a clear multiplying model: As to be expected, main effects were significant, $F(3, 27) = 231.97$ and 305.62, $p < .01$, for width and height, respectively. The significant overall interaction, $F(9, 81) = 24.87$, $p < .01$, was concentrated nearly exclusively (99%) in the bilinear component, $F(1, 9) = 221.79$, $p < .01$, whereas the residual part was not significant, $F(8, 72) = .26$.

Additional results. In the present experiment, the stimuli were constructed such that two subsets with special properties referring to an adding or a multiplying model could be selected for an additional statistical analysis:

On the one hand, there were three rec-

tangles with the same area (4 × 16, 8 × 8, and 16 × 4 cm), the sum of edge lengths of each of the two rectangles, however, being a fourth greater than that of the square (20 vs. 16 cm). The 5-year-olds judged these three rectangles as being different in area, $F(2, 18) = 7.06$, $p < .01$, whereas this does not hold true for the older age groups, $F(2, 18) = .22$, .26, and 1.93, $p > .10$, for the 8-year-olds, 11-year-olds, and adults, respectively (cf. Figure 2).

On the other hand, there were four rectangles with the same sum of edge lengths (4 × 16, 8 × 12, 12 × 8, and 16 × 4 cm), the two broader rectangles, however, being a half larger in area than the two narrower ones (96 vs. 64 cm^2). For this subset of stimuli, the situation is reversed: The 5-year-olds did not judge the four rectangles as being different in area, $F(3, 27) = .22$, whereas the 8-year-olds, 11-year-olds, and adults clearly did, $F(3, 27) = 5.15$, 16.82, and 82.56, $p < .01$, in the above order (cf. Figure 2).

As in Experiment 1, the additional analysis for the 5-year-olds yielded those results that an adding rule would predict. Further, the additional data found here suggest that the 8- and 11-year-olds, at least for these two subsets of rectangles, combined dimensions according to that multiplying rule adults used over the whole range of stimuli.

For a further analysis of the judgments in the above subsets of stimuli, the age variable was ignored, and adders and multipliers were separated out (cf. Table 1). The two groups of subjects performed quite differently in these special comparisons. The adders judged the three rectangles with the same area and different sum of edge lengths as being different in area, $F(2, 30) = 13.85$, $p < .01$, but not the four rectangles with the same sum of edge lengths and different area, $F(3, 45) = .31$. The reverse was true for the multipliers: They judged the latter subset as being different in area, $F(3, 54) = 78.55$, $p < .01$, but not the former one, $F(2, 36) = .01$. That is, adders as well as multipliers judged these special subsets as would be predicted from the general integration rule found for the whole range of stimuli. This result indicates that the rule, in the course of development, does not apply first to some subset and only later to the entire set.

Discussion

The hypothesis that young children combine stimulus dimensions according to an algebraic rule that is different from that used by adults was clearly supported by their judgments of rectangle area. Whereas the adults integrated width and height information multiplicatively, thereby replicating the finding of Anderson and Weiss (1971), the 5-year-olds combined the two dimensions according to an adding rule. Further, the highly significant main effects obtained in all age groups for the width and height factor show that children as well as adults based their judgments on both relevant stimulus dimensions.

The results of Experiment 2 show that there is no abrupt change in the course of development in the integration rule. The individual-subject analyses suggest, rather, that from 5 years to some time past 11 years, there is a gradual increase in the probability that a child will shift from an adding to a multiplying strategy. Furthermore, the analysis of the equal area and equal perimeter subsets of the stimuli indicates that this structure emerges full blown.

The adding model found here to describe the responses of the younger children has an interesting implication for the typical conservation task, in which an object with a narrower shape is compared with a wider one. It is usually found that the narrower one is overestimated by a nonconserver (e.g., Miller, Grabowski, & Heldmeyer, 1973). In his equilibration model for conservation development, Piaget (1970) explains this behavior via the concept of centering. He argues that a nonconserver attends only to one—usually the extended—dimension and disregards the other. The present results suggest another interpretation: It is not the centering on one dimension that causes nonconservation, but, rather, the fact that the relevant stimulus dimensions are added instead of multiplied.

It has to be pointed out that multiplication in the sense of the present study may

not be the same as logical multiplication that according to Piaget is necessary for conservation. Empirically, this discrepancy is shown by the fact that in Experiment 2 there were some children as old as 11 years who did not use a multiplying rule, but it would be rare to find children at this age failing to conserve substance, area, or any other relevant quantity. The difference between the two concepts of multiplication also becomes evident by considering Piaget's definition of logical multiplication. In contrast to the more familiar connotation of multiplication, Piaget defines the term as the simultaneous combination of two or more elements or relations. In the conservation paradigm, logical multiplication requires only that the two relevant asymmetrical relations (e.g., "wider than" and "higher than") are considered jointly and that the child sees the relations as compensatory; that is, no quantitative estimates are necessary. It should be noted that logical multiplication as defined by Piaget can only be successful in the specific situation in which two objects that are perceived as being identical have to be compared after some property (e.g., shape) of one of them has been transformed.

Multiplication as studied here is more general. Not asymmetrical relations (with only two possible levels on each dimension), but the subjective values of total dimensional extension have to be combined. In the present task, the easiest way to combine width and height multiplicatively was simply to follow the instruction and to apply the principle of repeated addition, which is often used in didactics as an illustration of multiplication. That is, the subjects had to imagine the area sliced up and the slices transported and relocated end to end. In fact, most of the "multipliers" reported doing so. It may be argued that the process underlying these judgments is not really one of multiplying but one of adding and therefore that the multiplying model has only an "as if" status (cf. Anderson, 1974a, p. 261).

But if the operation required for such a multiplication really is no more than additive in character, why then did the judgments of the younger children not follow the bilinear pattern? It is not reasonable to assume that the children would have difficulties in adding slices, once they had imagined the area sliced up. The problem therefore seems to be located in the first stage of the judgment process. That is, the child may have difficulties in distributing his attention homogeneously over the whole area and therefore be unable to imagine the whole figure divided into equivalent parts, either (a) into components (e.g., vertical or horizontal slices) that vary in size and/or number across the different stimuli or (b) into components (e.g., small squares) of equal size and variable number. It is also possible that the child after having done one or the other successfully fails when trying mentally to transport the parts of the area to the response scale.

Two mechanisms can be thought of to account for the adding rule of the children. The first is the width + height rule. By this rule, the child merely examines width and height and adds the subjective values. The second mechanism may be described as the perimeter rule. Using this rule, the child would scan the outline border of the rectangle, and the response would be based on an estimate of the total perimeter, that is, $2 \times$ (width + height). The latter seems more probable, mainly for two reasons. First, in both experiments of the present study, the perimeter rule provides a better fit for the data than the width + height rule. Although the subjective values need not conform to the actual values, it is impressive that the responses of the "adders" coincide with the objective perimeters rather closely. Second, as has been shown in studies on formal reasoning (e.g., Aebli, 1951), children as old as 12 years tend to accept the perimeter of a rectangle as a valid measure for its area. This is not as senseless as one may think because in everyday life perimeter correlates quite well with area and thus can be applied successfully as a basis for area judgments in many situations.

Although the perimeter rule provides a plausible explanation of the present findings, it may be possible that the additive judgment pattern found here is one manifestation of a general adding rule that

children use not only in judging rectangle area but in a variety of other situations as well. This idea of a general purpose adding-type rule, which recently has been suggested by Anderson and Cuneo (1978), is especially supported by two studies using dimensions that are not geometrical. In both of them, the response patterns of the children differed characteristically from those of the adults, who combined the dimensions as expected according to a multiplying rule. In the study of Kun et al. (1974), young children combined ability and effort information additively to predict performance. In a more recent study by Wilkening and Becker (Note 1), 5- and 6-year-olds had to predict distance on the basis of information about velocity and time. Again, an adding model was found to be the best fit for the data. Future research is needed to determine whether the surprising similarity between the findings of the different studies is only analogous from a formal point of view or if there are, in fact, some general psychological processes that can account for the development from an adding to a multiplying information integration strategy (cf. Wilkening, in press).

Reference Note

1. Wilkening, F., & Becker, J. *Development of dimensional integration in children's perceptual judgments: Some experiments with area, volume, and velocity*. Paper presented at the conference entitled "Structure and Process Models in the Study of Dimensionality of Children's Judgments," Kassel, Germany, June 1977.

References

Aebli, H. *Didactique psychologique: Application à la didactique de la psychologie de Jean Piaget*. Neuchâtel, Switzerland: Delachaux et Niestlé, 1951.

Anderson, N. H. Algebraic models in perception. In E. C. Carterette & M. P. Friedman (Eds.), *Handbook of perception* (Vol. 2). New York: Academic Press, 1974. (a)

Anderson, N. H. Cognitive algebra. In L. Berkowitz (Ed.), *Advances in experimental social psychology* (Vol. 7). New York: Academic Press, 1974. (b)

Anderson, N. H., & Cuneo, D. O. The height + width rule in children's judgments of quantity. *Journal of Experimental Psychology: General*, 1978, *107*, 335–378.

Anderson, N. H., & Weiss, D. J. Test of a multiplying model for estimated area of rectangles. *American Journal of Psychology*, 1971, *84*, 543–548.

Butzin, C. A., & Anderson, N. H. Functional measurement of children's judgments. *Child Development*, 1973, *44*, 529–537.

Graesser, C. C., & Anderson, N. H. Cognitive algebra of the equation: Gift size = generosity × income. *Journal of Experimental Psychology*, 1974, *103*, 692–699.

Hendrick, C., Franz, C. M., & Hoving, K. L. How do children form impressions of persons? They average. *Memory & Cognition*, 1975, *3*, 325–328.

Kempler, B. Stimulus correlates of area judgments: A psychophysical developmental study. *Developmental Psychology*, 1971, *4*, 158–163.

Kosslyn, S. M., Pick, H. L., Jr., & Fariello, G. R. Cognitive maps in children and men. *Child Development*, 1974, *45*, 707–716.

Kun, A., Parsons, J. E., & Ruble, D. N. Development of integration processes using ability and effort information to predict outcome. *Developmental Psychology*, 1974, *10*, 721–732.

Miller, P. H., Grabowski, T. L., & Heldmeyer, K. H. The role of stimulus dimensions in the conservation of substance. *Child Development*, 1973, *44*, 646–650.

Odom, R. D., & Corbin, D. W. Perceptual salience and children's multidimensional problem solving. *Child Development*, 1973, *44*, 425–432.

Piaget, J. Piaget's theory. In P. H. Mussen (Ed.), *Carmichael's manual of child psychology* (Vol. 1). New York: Wiley, 1970.

Piaget, J., & Inhelder, B. *Le développement des quantités physiques chez l'enfant*. Neuchâtel, Switzerland: Delachaux et Niestlé, 1941.

Piaget, J., & Inhelder, B. *The child's conception of space*. London: Routledge and Kegan Paul, 1956.

Piaget, J., Inhelder, B., & Szeminska, A. *The child's conception of geometry*. New York: Basic Books, 1960.

Shepp, B. E., & Swartz, K. B. Selective attention and the processing of integral and nonintegral dimensions: A developmental study. *Journal of Experimental Child Psychology*, 1976, *22*, 73–85.

Singh, R., Sidana, U. R., & Saluja, S. K. Playgroup attractiveness studied with information integration theory. *Journal of Experimental Child Psychology*, 1978, *25*, 429–436.

Singh, R., Sidana, U. R., & Srivastava, P. Averaging processes in children's judgment of happiness. *Journal of Social Psychology*, 1978, *104*, 123–132.

Wilkening, F. Development of dimensional integration in children's perceptual judgment: Experiments with area, volume, and velocity. In F. Wilkening, J. Becker, & T. Trabasso (Eds.), *Information integration by children*. Hillsdale, N.J.: Erlbaum, in press.

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